

SECTION 271000

STRUCTURED CABLING

PART 1 - GENERAL

1.1 DESCRIPTION:

- A. The Work shall consist of furnishing and installation of voice and data structured communication cabling as shown on the drawings and specified herein.

1.2 REFERENCES:

A. SASO - Saudi Arabian Standards Organization.

SASO 1357 Electrical installation of buildings – Part 7: Requirements for special installation or locations – Section 707: Earthing Requirements for the installation of data processing equipment.

B. ISO – International Standards Organization

ISO/IEC 11801 Information Technology – Generic Cabling for Customer Premises

Comply

ISO/IEC TR 14763-1 Information technology – Implementation and operation of customer premises cabling – Part 1: Administration

ISO/IEC TR 24750 Information Technology – Assessment of installed balanced cabling channels in order to support 10 GBASE – T.

C. IEC - International Electrotechnical Commission

IEC 60603-7 Connectors for Electronic Equipment – Part 7-1 & 7-7

Comply

IEC 60793 All parts relevant to these Specifications

IEC 60794 All parts relevant to these Specifications

Comply

IEC 61156-1 to 6 Multicore and symmetrical pair/quad cables for digital communications – Part 1 to 6

Comply

IEC 61935-1 & 2 Generic cabling systems – Specification for the testing of balanced communication cabling in accordance with ISO/IEC 11801 – Part 1 & 2

Comply

IEC 60332 Test on electric cables under fire conditions

Comply

IEC 60754 Tests on Gases Evolved During Combustion of Material from Cables

Comply

D. EIA - Electronic Industries Association

RS-453	Dimensional, Mechanical and Electrical Characteristics Deferring Phone Plugs and Jacks	
EIA 310	Racks, Panels and Associated Equipment	
EIA/TIA 492AAAA	Detail Specification for 62.5-UM Core Diameter/125-UM Cladding Diameter Class 1A Graded-Index Multimode Optical Fibers	
EIA/TIA 526-14-A	Optical Power Loss Measurements of Installed Multimode Fiber Cable plant - OFSTP-14a.	
EIA/TIA 568 A	Commercial Building Telecommunications Wiring Standard	Comply
EIA/TIA 568-B.1	Commercial Building Telecommunications Cabling Standard, Part 1: General Requirements (ANSI)	Comply
EIA/TIA 568-B.2	Commercial Building Telecommunications Cabling Standard, Part 2: Balanced Twisted-Pair Cabling Components (ANSI)	Comply
EIA/TIA 568-B.3	Optical Fiber Cabling Components (ANSI)	Comply
EIA/TIA 569	Commercial Building Standard for Telecommunications Pathways and Spaces	
EIA/TIA 606-A	Administration Standard for the Telecommunications Infrastructure of Commercial Buildings	
EIA/TIA 607	Commercial Building Grounding/Bonding/Requirement Standard	
EIA/TIA TSB 67	Transmission Performance Specifications for Field Testing of UTP Cabling Systems	
EIA/TIA TSB 155	Additional Guidance for 4 – Pair 100 ohms Category 6 Cabling for 10 GBASE-T Application	
E.	<u>IEEE - Institute of Electrical and Electronic Engineers</u>	
IEEE 802.3 AN	10 GBASE – T Standards 2006	Comply
F.	<u>ICEA - Insulated Cable Engineers Association</u>	
ICEA S-80-576	Category 1 & 2 Individually Unshielded Twisted Pair Indoor Cables for Use in Communications Wiring Systems	
ICEA S-83-596	Standard for Fiber Optic Premises Distribution Cable	
G.	<u>CFR - Code of Federal Regulations</u>	

47 CFR 68 Connection of Terminal Equipment to the Telephone Network

H. NFPA - National Fire Protection Association

NFPA 70 National Electric Code

NFPA 78 Lightning Protection Code

I. NEMA - National Electric Manufacturer Association

NEMA 250 Enclosures for Electrical Equipment **Comply**

NEMA WD 6 Wiring Device - Dimensional Requirements

J. UL - Underwriters Laboratories Inc. **Comply as Noted EN Standards**

UL 50 Enclosures for Electrical Equipment

UL 1863 Standard for Safety, Communications Circuit Accessories

K. BICSI TDMM Manual (Latest edition)

1.3 SUBMITTALS:

A. General: Submit the necessary complete sets of documentation indicating type, size, rating, style, catalog number, Manufacturers names, photographs and / or catalog data sheets for all items to ensure compliance with Specifications. This documentation shall be subject to the approval of the Owner Representative and no equipment shall be ordered without his approval for all equipment and devices, which are shown on documents (drawings, BOQ, etc). Any document submitted by the contractor during the tender stage shall not be considered as technical submittal, it shall be used only to approve the manufacture name. During technical submittal stage, contractor shall submit all required technical document for study and approval. This study may drive to reject a vendor had been previously approved during the tender stage. **Noted**

B. Product Data: Submit manufacturer's technical product data, including:

1. Compliance sheet to the specification with cross reference to related items in data sheet, point by point, indicating deviations, if any, with reasons for such deviations, also indicate any extra features / specifications. **Noted**
2. Complete one-line riser diagram(s) showing all system components. **Noted**
3. Pin to pin riser diagram showing all system components and interconnection between the system and all other related systems. **Noted**
4. Complete description data including UL listing or any equivalent standards for all system components.

5. Provide all system related calculations supported by manufacturer specific software and perform all necessary calculations validating the shop drawing system distribution and related installations.
6. Complete description of system operations and functions of each system component. **Noted**
7. Manufacturer's technical product data sheets, including quantities and specifications of every item, specification of every device, system component, quantities, software and installations for each unit of equipment. **Noted**
8. Contractor shall submit a list of the Manufacturer's authorized, local representative responsible for installation coordination and service, pre-qualification for the system provider and installer. **Noted**
9. Complete description and data including related standards for all system components. **Noted**
10. Documents indicating system expandability options included in the submitted products and system included redundancy features. **Noted**
11. Documents indicating that the supplied system components are the latest and most updated products from the system manufacturer and software versions are the latest produced by the system manufacturer. **Noted**
12. Integration documents approved and stamped by overall project systems integrator. Such documents should illustrate that the system will satisfy the project system integrator requirements in terms of software interface protocols and hardware interfaces to achieve fully interoperable systems. **Noted**
13. All technical material submittals must be provided in both hard copy and soft copy. The softcopy must allow for searching capabilities. The technical material submittal will be rejected if the softcopy was not included. **Noted**
14. Testing & commissioning procedures according to system manufacturer requirements and calibration certificate for testing equipment **Noted**
15. Configuration plan including all management plans to be implemented according to project operational team. **Noted**
16. Provide Bandwidth calculation sheets for all switches in the network. **Noted**
17. Verify and submit power consumption and heat dissipation calculation sheets utilizing system manufacturer software and make all necessary changes and adaptation on site without extra cost. **Noted**
18. Submit Wi Fi Coverage patterns and accordingly modify, add, relocate wi fi outlets to provide full coverage of the building enabling VOIP over Wi Fi network in all building areas. **Noted**

- C. Shop Drawings: Provide shop drawings & Composite drawings showing equipment, device locations, labeling, part number and connecting wiring of the systems, including riser diagrams, rack elevations. Shop drawings shall include, but not be limited to the following: **Noted**
1. Complete pin to pin one-line riser diagram(s) showing all approved equipment, size, type and number of all conductors, interconnection between the system and all other related systems. **Noted**
 2. Installation details for all system components. Installation details drawings shall show all accessories used in installation such as back boxes, glands, washers, etc. **Noted**
 3. Complete Grounding details as per system manufacturer requirements. Indicating connections with dedicated grounding system including connections with telecommunication grounding bus bars as part of overall grounding and bonding system including telecom main grounding bus bar **Shop Drawings Contractor Scope**
 4. Exact location of every single component of the system in coordination with all other works (Electromechanical, Arch., and Civil). **Noted**
 5. Colored Composite drawings for all MEP systems showing all coordinated MEP works.
 6. Signature of all involved contractor Owner Representatives representing different disciplines (Electromechanical, Arch, and Civil etc) to confirm their approval of such locations, installations, and coordination. **Noted**
 7. Complete sequence of operations and functions of the system. **Noted**
 8. All drawings should be submitted in both softcopy and hardcopy formats. No of copies shall be according to contract documents requirements (at least three copies shall be submitted)
- D. System supplier & Installer qualifications documents: System supplier & Installer should submit the required certifications indicating that the system supplier and installer company is certified from system manufacturer to perform all system supplying, installations, testing and commissioning supported by system installers CVs and related manufacturer certifications. The system supplier should be involved in a strong relationship with system manufacturer with proven reference to similar projects. The system supplier & installer should submit reference of similar projects and abide by all requirements in project documents (specifications, drawings, BOQ, and method of measurements) **Noted**
- E. Manuals: Submit complete manufacturer Installations, maintenance and operation manuals including spare parts list for each system component, including furnished specialties and accessories. Include this data, product data, and shop drawings in the manuals in accordance with other relevant documentation. **Noted**
- F. Composite Coordinated drawings illustrating all MEP devices and equipment. Contractor shall validate all system components power requirements and

indicate on drawings that all system power requirements have been coordinated with electrical team. **Noted**

- G. Documents indicating the working environment according to system manufacturer regarding temperature and humidity are achieved and coordinated with HVAC team. **Noted**
- H. Finalized Reflected ceiling drawings illustrating all works in the ceilings after finalizing the coordination process. **Noted**
- I. Testing & commissioning plan according to manufacturer testing procedures. Third party testing agency shall be nominated for the whole project. **Noted**
- J. Wiring Diagrams: Show typical wiring schematics including workstation outlets, jack and jack assemblies, patch cords, patch panels, fiber-optic boxes and other equipment. **Noted**
- K. Samples: For workstation outlets (TO), jacks, jack assemblies, and faceplates for color selection and evaluation of technical features. **Noted**

1.4 TRANSPORTATION, HANDLING AND STORAGE:

- A. Deliver equipment and components in factory-fabricated containers or wrappings, which properly protect equipment from damage. **Noted**
- B. Store equipment and components in original packaging. Store inside in a well-ventilated space protected from weather, moisture, soiling, humidity, and extreme temperatures. **Noted**
- C. Handle equipment and components carefully to prevent damage, breaking, and scoring of finishes. Do not install damaged units or components; replace with new. **Noted**

1.5 WARRANTY:

- A. The manufacturer must guarantee to the End User that the products referenced within the specific Warranty Modules (Class E System) when correctly installed in accordance with installation guidelines:
 - 1. Will be free from product defects in materials and workmanship. **Noted**
 - 2. Are guaranteed to exceed the Class E Channel and Permanent Link requirements as specified in ISO/IEC 11801:2002 **Noted**
 - 3. Supports the following application (not limited):
 - 10BASE T Ethernet **Noted**
 - 100BASE T Fast Ethernet
 - 1000BASE TX Gigabit Ethernet
 - 10GBASE-T
 - 155Mbit ATM
 - 1000Mbit ATM (CB1G)

- 10 GBASE-T

4. For a duration of 20 years

- B. All components including the patch cords have to be produced by the same cabling system manufacturer to ensure warranted performances and applications against the standards. **Noted**

1.6 QUALITY ASSURANCE:

- A. Manufacturer's Qualifications: The items provided under this contract will be from manufacturers that have a minimum of 5 years' experience in producing the types of systems and equipment specified. **Noted**
- B. Installer Qualifications: Specialist subcontractor with at least 5 years of successful installation experience with projects utilizing data system similar to that required for this project. Subcontractor shall be subject to approval of Engineer. **Noted**
- C. Materials and installation shall comply with the specified Codes and Standards. **Noted**
- D. Single Source Responsibility: All components and accessories shall be product of single manufacturer except for cables. **Noted**

1.7 ENVIRONMENTAL REQUIREMENTS:

- A. Connecting hardware shall be rated for operation under ambient conditions of 0 to 60 degrees C and in the range of 0 to 95 percent relative humidity, non-condensing. **Noted**

PART 2 - PRODUCTS

2.1 SYSTEM DESCRIPTION:

- A. The voice and data communication cabling shall consist of horizontal and backbone cables and connecting hardware to transport data (including LAN/WAN) and voice (telephone) system signals between related as shown on drawings. Numbers of outlets shall be as shown on drawings and documents. The Data Local Area Network (LAN) Sub-Contractor shall coordinate his work with telephone equipment Sub-Contractor. **Noted**
- B. The system shall support CATEGORY 6A 10GBASE -T **Comply**
- C. The Data Network will perform all switching and routing functions for voice, video and data services and applications. The network shall serve all IP based services such as VOIP, IP CCTV, IPTV, etc., as applicable. **Comply**
- D. The Core Switches will be located at the heart of the network and shall provide very fast switching, intelligent high-performance platform for deploying numerous concurrent intelligent services without degrading the overall performance of the network. The core switches shall perform various functions using various service modules for wan interfaces, security firewalls...etc. Core Switches should support MPLS and act as P-Routers. **Comply**

- E. Connectivity between DC Switches and Core Switches should be based on 40G. **Comply**
- F. Connectivity between Core Switches and Distribution Switches should be based on 40G. **Comply**
- G. The distribution switches will be used to provide aggregation for various services and filter traffic that will reach the network core. Each distribution switch shall be connected to each core switch via 40GbE link and the edge/Access switch will be connected to the distribution switch via 2x10GbE. The connections between each edge switch and the distribution switch must be on a different card on the distribution switch. **Comply**
- H. Access Switch: Access Switches will be located at the wiring closets in communication rooms; they will provide users with access to the network. The Access switches shall support power over Ethernet POE+ Delivering 30Watt on all ports. Each Access switch shall be connected to distribution switch via 2x 10 GbE. each link shall be on different module in the distribution switch. **Comply with note
Quoted switches are Fixed**
- I. Distribution and Access Switches should support MPLS and act as PE-Routers. **Comply with Deviation
MPLS supported on Core/Dist. Not standard or applicable for Access switches**
- J. Connectivity of DC Switches to Servers and Storage shall be based on FCOE - allowing convergence of SAN and LAN. **Not Comply**
- K. Network Equipment Management Software: The management software's main function is to enable remote and centralized configuration of the different parameters and functions of switches and routers of the Data Network. **Comply But Not Included In BOQ**
- L. Access Switch Stack should consist of maximum 4 Switches and have stacking bandwidth of 64 Gbps. (The current design does not consider implementing any stacking techniques; however, the provided switches must support stacking whenever is required by the operator team). **Comply**
- M. Contractor is fully responsible to determine the number of distribution and core switches chassis to accommodate the required number of uplinks between various layers according to the network riser diagram. This shall be determined according to the maximum port number that is supported by the chassis which is defined by the switching network manufacturer. **Noted but fixed core switches are quoted**
- N. Horizontal cable and its connecting hardware provide the means of transporting signals between the telecommunications outlet/connector and the horizontal cross-connect located in the communications equipment room. This cabling and its connecting hardware are called "permanent link," a term that is used in the testing protocols.
 - 1. TIA/EIA-568-B.1 requires that a minimum of two telecommunications outlet/connectors be installed for each work area. **Comply**
 - 2. Horizontal cabling shall contain no more than one transition point or consolidation point between the horizontal cross-connect and the telecommunications outlet/connector.
 - 3. Bridged taps and splices shall not be installed in the horizontal cabling. **Comply**
 - 4. Splitters shall not be installed as part of the optical fiber cabling.

- O. A work area is approximately. 9.3 sq. m and includes the components that extend from the telecommunications outlet/connectors to the station equipment. **Noted**
- P. The maximum allowable horizontal cable length is 90 m. This maximum allowable length does not include an allowance for the length of 4.9 m to the workstation equipment. The maximum allowable length does not include an allowance for the length of 4.9 m in the horizontal cross-connect. **Noted**
- Q. Backbone cabling system shall provide interconnections between communications equipment rooms, main terminal space, and entrance facilities in the telecommunications cabling system structure. Cabling system consists of backbone cables, intermediate and main cross-connections mechanical terminations, and patch cords or jumpers used for backbone-to-backbone cross-connection. **Noted**
- R. Backbone cabling cross-connections may be located in communications equipment rooms or at entrance facilities. Bridged taps and splitters shall not be used as part of backbone cabling. **Noted**

2.2 GENERAL:

- A. Materials and equipment shall be the standard products of a manufacturer regularly engaged in the manufacture of the products and shall be of latest technology/version available at the time of installation and has been in satisfactory use for at least 1 year prior to installation. Materials and equipment shall conform to the respective publications and standards specified herein and to the applicable requirements of ISO/IEC 11801,2002 EIA/TIA 568 and to all additional requirements of IEEE 802.3 AN and TIA TSB – 155 and ISO/IEC TR 24750 to support the 10 GBASE-T application over 100 meters channel. **Noted**

2.3 HORIZONTAL CABLING:

- A. Horizontal Distribution Cable: **Comply with note
Datwyler Cat 6A fully supports 10GBASE-T & ISO/TIA standards**
 - 1. Horizontal distribution cable shall be 10 G (UTP) Category 6A. **Comply as F/FTP**
 - 2. The cable shall be a four, thermoplastic-insulated, individually twisted pairs of copper conductors; No.23 AWG, color-coded; enclosed in PVC jacket or a material that does not give off toxic fumes (Low Smoke Zero Halogen - LSZH) in case of fire and offer flame propagation retardant properties. **Comply as F/FTP**
 - 3. Cable shall be designed to support the IEEE 802.3 1000Base.T (Gigabit Ethernet), potentially 1.2 Gbps ATM and 2.4 Gbps ATM as well as analogue broadband video up to 550 MHz. **Comply**
 - 4. Cable shall meet or exceed attenuation and NEXT requirements in ISO/IEC 11801, CENELEC EN50173 and EIA/TIA 568A. **Comply**
 - 5. All pairs must have impedance of 100 Ohms, with a tolerance of +/- 15 Ohms. In the construction of the cable crosstalk performance shall be maintained using a C3 (Central dielectric Crosstalk Cancellation)

member set between the 4 pairs.

6. Insulators in standard Blue/White, Orange/White, Green/White, Brown/White colors must cover the conductors. **Comply**
7. Cable shall be of latest technology/speed manufactured and available at the installation time. **Comply**
8. Cable shall be used for horizontal run between data, voice (telephone), clock and security system outlets and floor communicable cabinet. **Comply**
9. Available in both 305m, 500m and 1000m reels **Comply**
10. Meet the following electrical characteristics: **Comply**

Max DC Resistance (@ 20°C)	<8.5Ω /100m
Characteristic Impedance (no impedance averaging allowed)	1-100 MHz: 100 ohms ± 15% 100 - 750MHz: 100 ohms ± 22%
Nominal Velocity of Propagation (NVP)	LSOH – 67%

11. Provide the following 100m, 4-connector topology performance (std ref. values in grey cells): **Comply**

Frequency (MHz)	Insertion Loss (dB/100 m)		NEXT Loss (dB/100 m)		PS NEXT Loss (dB/100 m)		ACR-F (dB/100 m)		PS ACR-F (dB/100 m)		Return Loss (ns/100 m)		Propagation Delay (ns/100 m)		Delay Skew	
1.0	2.0	1.8	74.3	86.0	72.3	82.3	67.8	91.0	64.8	85.0	20.0	33.0	570.0	545.0	45.0	30.0
4.0	3.8	3.4	65.3	77.0	63.3	73.3	55.8	79.0	52.8	73.0	23.0	35.5	552.0	527.0	45.0	30.0
10.0	6.0	5.4	59.3	71.0	57.3	67.3	47.8	71.0	44.8	65.0	25.0	38.0	545.0	520.0	45.0	30.0
16.0	7.6	6.9	56.2	68.0	54.2	64.2	43.7	67.0	40.7	61.0	25.0	35.2	543.0	518.0	45.0	30.0
20.0	8.5	7.7	54.8	67.0	52.8	62.8	41.8	65.0	38.8	59.0	25.0	35.0	542.0	517.0	45.0	30.0
31.25	10.7	9.9	51.9	64.0	49.9	59.9	37.9	61.0	34.9	55.0	23.6	33.1	540.0	515.0	45.0	30.0
62.5	15.4	14.3	47.4	59.0	45.4	55.4	31.9	55.0	28.9	49.0	21.5	32.2	539.0	514.0	45.0	30.0
100.0	19.8	18.1	44.3	56.0	42.3	52.0	27.8	51.0	24.8	45.0	20.1	31.6	538.0	513.0	45.0	30.0
160.0	25.6	23.2	41.2	53.0	39.2	49.2	23.7	47.0	20.7	41.0	18.7	30.1	537.0	512.0	45.0	30.0
200.0	29.0	27.3	39.8	52.0	37.8	47.8	21.8	45.0	18.8	39.0	18.0	29.8	537.0	512.0	45.0	30.0
250.0	32.8	31.1	38.3	50.0	36.3	46.0	19.8	43.0	16.8	37.0	17.3	28.7	536.0	511.0	45.0	30.0
300.0*	36.4	35.0	37.1	49.0	35.1	45.0	18.3	38.0	15.3	35.0	16.8	28.0	536.0	511.0	45.0	30.0
400.0*	43.0	40.0	35.3	47.0	33.3	43.0	15.8	36.0	12.8	33.0	15.9	27.1	536.0	511.0	45.0	30.0
500.0*	48.9	42.0	33.8	47.0	31.8	42.0	13.8	34.0	10.8	32.0	15.2	26.0	536.0	510.0	45.0	30.0
550.0*	51.8	43.0	33.2	46.0	31.2	42.0	13.0	33.0	10.0	31.0	14.9	26.0	536.0	510.0	45.0	30.0
625.0*	55.8	44.9	32.4	46.0	30.4	41.0	11.9	33.0	8.9	29.0	14.5	25.0	535.0	505.0	45.0	25.0
750.0*	62.3	49.0	31.2	45.0	29.2	41.0	10.3	32.0	7.3	27.0	14.0	25.0	535.0	504.0	45.0	25.0

- B. Category 6 Twisted Pair Data Cables Unshielded Twisted Pair (UTP) LAN Grade, IEEE 802.3, 10 BASE-T, EIA/TIA-568, 10 & 16 Mbps, 24 AWG solid bare annealed copper. **Comply**
- C. Telecom Outlets (TO): All 500 MHz category 6A information outlets designed for termination of 4-pair balanced twisted-pair category 6A copper cable must possess the following characteristics at the minimum: **Comply**
 1. Exceed category 6A component compliance through the frequency range of 1 to 250MHz with usable bandwidth to 500MHz. **Comply**

2. Telecom outlets shall be CAT 6A (UTP) RJ45. **Comply**
 3. Each connector shall provide both T568A and T568B color code identification for the pins at the rear of the connector. **Comply**
 4. The punch down is to be in accordance with the T568B color code. **Comply**
 5. Reassignment of pairs is forbidden. **Comply**
 6. All conductors from the CAT6A 4 pair cable shall be terminated on the respective contacts. **Comply**
 7. Universal design allows the same outlet to be mounted in flat or angled orientation. **Comply**
 8. Have available termination tools. **Comply**
 9. Be backwards compatible to allow lower performing categories of cables or connecting hardware to operate to their full capacity. **Comply**
 10. Allow installation from the front or rear of the faceplate and allow for the jack to pass through the faceplate without re-termination. **Noted**
 11. Provide color-coded, snap-in icons available for circuit identification. **Noted**
 12. To avoid installation errors, the wire organizer of the snap-in connector must be identified by the same standard color coding as the wires. **Noted**
 13. All CAT 6A (UTP) RJ45 connectors shall be fully compliant with the ISO/IEC 11801:2002 standard. The presentation of the outlet shall provide for labeling and identification. **Comply**
 14. A transparent window shall protect the labeling tag. **Comply**
 15. All outlets shall be equipped with shutters. **Comply**
 16. Snap-in Format (Where required): The dimensions of the Snap-in format connectors shall be H x W x D: 19.5 mm x 16.8 mm x 35.7 mm. The same format connectors (unscreened) shall be used on each link. The connector fits in specific structural hardware for Snap-in format of third parties. If not available, the Snap-in connector can be used in combination with a keystone clip and specific structural hardware for keystone connectors. **Quoted modules are Shielded (superior to UTP). Complies with T568A/B, shutters, and snap-in keystoneformat.**
- D. Faceplates: All faceplates installed, as part of this specification shall have these minimum features listed below: **Comply with Note**
Single-port angled faceplates are quoted
1. Be applicable to both fiber and copper applications. **Comply With note for copper**
 2. Be available in 2 and 4 port single-gang configurations. **Single Port Proposed**
 3. Allow modules to be removed from the front of the faceplate. **Comply**
 4. Allow UTP modules to pass through faceplates even after termination. **Comply**

5. Have write on designation labels for circuit identification together with a clear plastic cover. **Noted**
 6. Feature easily removable designation label covers which can be removed without use of tools. **Noted**
 7. Have as a minimum the standard colors of white, bright white, and alpine white. **Noted**
 8. Have optional modular furniture adapters available.
 9. Have surface mount boxes single gang faceplates. **Comply**
 10. Be manufactured using UV resistant, high impact thermoplastic to prevent color fading and provide additional durability. **Comply**
- E. Copper Patch Panels (RJ45): **Comply with note: Proposed panels are Shielded,**
1. Patch panels must have 19" equipment practice dimensions to permit mounting in standard cabinets, racks or bays. **Comply**
 2. Provide 24 ports Snap-in format (Modular) patch panels equipped with a cable management mechanism that provides strain relief, earthing and grounding features. **Comply**
 3. The presentation of the Patch Panel shall provide for labeling using a printed numbering system. **Comply**
 4. The connector shall provide both T568A and T568B color code identification for the pins at the rear of the connector. **Comply**
 5. Patch panel outlets must have each Category 6A (UTP) RJ45 connector, connected separately. **Comply**
 6. The punch down is to be in accordance with the T568B color code. Reassignment of pairs is forbidden. **Comply**
 7. The screened connectors from modular patch panel shall provide termination facilities for the drain wire of each category 6A UTP cable. **Comply**
 8. All conductors from the 4 pair cable are to be terminated on the respective contacts. To avoid installation errors, IDC blocks must be identified by the same standard color coding as the wires. **Comply**
 9. Each patch panel shall provide a means to locate and clamp the incoming cables without causing damage to the cables or affecting the performance of link. **Noted**
 10. The installer must avoid any risk of cable pinching or compression during the installation of cable. Therefore, use of Velcro cables is preferred. **Noted**
 11. In the rack, the Patch Panels shall be separated by metallic patch-

guides that have a closed front to protect the patch cords. The height of these guides will be 1U or 2U depending on the layout of the rack. **Noted**

12. The metal frame of patch panel must not be earthed to the cabinet with a separate earth strap if the patch panels automatically make contact with the metal frame of the cabinets. If the cabinet is not designed to provide the panels with automatic earth, then the patch panels must be connected with separate earth strap to the earth key. **Noted**

Deviation

F. Intelligent Patch Panels: **standard conventional patch panels are proposed exactly as per the provided BOQ description**

1. Intelligent physical infrastructure management by providing a modular and scalable approach to meet the demands of enterprise and data center installations.
2. Standard copper and fiber cabling is used for all terminations onto the Single-port angled faceplates are quoted back of these panels while Patch Cords are used for cross-connecting or interconnect installations.
3. Mount to standard EIA 19" rack or 23" racks with extender bracket.
4. Accommodates intelligent modules that mount in the rear and utilize no additional rack space zero RU.
5. Accepts all UTP, STP, and LC Fiber Optic Modules.
6. Accepts an Interface Unit to provide patch cord tracing and provisioning of switch ports with Interconnect Patch Cords or patch field mapping with Cross-Connect Patch Cords.

G. Patch Cords: To achieve a Class E 10 G Channel performance all Patch Cords will be Category 6A rated. All patch cord cable will be made from PVC or Low Smoke Zero Halogen (LSZH) material (when used in plenum areas).

1. Augmented category 6 component compliant out to 250MHz with operational bandwidth to 500MHz. **Comply**
2. Factory assembled and 100% transmission tested with laboratory grade network analyzers for proper performance up to 500MHz. **Comply**
3. Backwards compatible with lower performing categories. **Comply**
4. Equipped with identical modular 8-position plugs on both ends, wired straight through with standards compliant wiring. **Comply**
5. Have full 360° shield coverage and metal plug housing to provide durability and resistance to damage. **Comply**
6. Have a PCB based plug to enable high levels of performance? **Comply**
7. Have fixed position front contacts to ensure plug quality and consistent mating with outlets. **Comply**
8. Have internal rear contacts to maintain cable pair symmetry to the

point of termination. **Comply**

9. Have a boot that features an ultra slim design for high density applications and snag free operation. **Comply**
10. Use modular plugs, which exceed FCC CFR 47-part 68 subpart F and IEC 60603-7 specifications and have 50 micro-inches minimum of gold plating over nickel contacts. **Comply**
11. Have a dual jack construction for excellent alien crosstalk performance. **Comply**
12. Available in standard lengths of 3, 5, 7, 10, 15 and 20 ft. with custom lengths available upon request. **Comply**
13. Have optional colored clips to allow field color coding even when cords are already installed. Available in Black, White, Red, Gray, Yellow, Blue, Green and Orange **Comply**

H. Consolidation point/Zone Unit Enclosure: A zone unit is a termination connection point between open office cabling and horizontal cabling to allow for reconfiguration of the open office cabling. A zone unit is not a splice, it is considered to be a piece of connecting hardware. Installation guidelines are as follows: **Not Applicable / Out of Scope**

1. No more than one zone unit may be used in a single horizontal cable run.
2. Each horizontal distribution cable exiting the zone unit shall have all pairs terminated in all eight positions of the modular outlets in the work area.
3. Cross-connections or active equipment shall not be permitted at a zone unit.

The zone unit shall have the following characteristics as minimum:

1. Enclosure designed to accommodate 24- or 48- work area modules.
2. Constructed of 16-gauge steel.
3. Feature 1.8" x 4.1" cable access entry points.
4. Incorporate ¼-turn hook and loop managers for routing of cables entering/exiting the enclosure.
5. Have two-piece, base and cover design.
6. Feature a low-profile design, which is 1.8", or lower in height.

I. Class E link or channel: Manufacturer should demonstrate guaranteed minimum worst-case performance to compliant with class E channel performance according to the ISO/IEC 11801: 2002 standard. Components used must be compliant with the category 6A standard mentioned above and the manufacturer

should be able to demonstrate independent Delta verification. The performance of both the components and the link and channel should show stable performance up to 500 MHz in order to allow for possible future applications requiring crosstalk cancellation up to 500 MHz.

The system supplier must be able to demonstrate in-house design and manufacturing expertise for all components used (e.g., cables, outlets, panels and cords) in order to ensure compatibility of system elements.

2.4 BACKBONE CABLING: **Comply**

A. Data Backbone: This backbone will link the Switches located in the Floor Distributors (FD) to the Data network server through the Building Distributor (BD).

1. Optical Fiber Cable: The cable shall be suitable for connector manufacturer termination processes (LC, SC or ST connectors). Fiber splicing method should be used for termination of Fiber Cables using fiber pigtails manufactured by the same vendor. **Comply**

Construction: Tight buffered with water blocking Aramid/Glass yarn reinforcements or tape and shall be suitable for indoor or outdoor use making it ideal for short campus links without the use of transition joints. The cable strength member shall be glass yarn laid longitudinally between the fibers and the inside wall of the outer jacket. The cable shall be dielectric construction, i.e., with no metallic content. The cable shall be a dry construction i.e., with no gel content. The jacket material shall be waterproof LSZH with a minimum fire performance of IEC 332-part 3C.

Comply

2. Multimode Fiber Optic Cable:

- a. The Cabling system must be designed and installed according to ISO 11801 OM2 & OM3 compliant and should also meet EN50173 2nd editions and TIA/EIA 568B3. The Cable shall have 4, 6, 12, 24, 48, 72 cores, 50/125 µm universal distribution cable with improved performance.

Comply

- b. Low Smoke Zero Halogen - LSZH Jacket that does not give off toxic fumes in case of fire and offer flame propagation retardant properties.

Comply

- c. Gigabit Ethernet should be supported up to a minimum distance of 550 m. If required, extended distance option with laser optimization should be available to increase this distance up to 800 m in case of 850 nm fiber and 2,000 m in case of 1300 nm.

Comply

- d. In case 10G Ethernet application has to be used, the fiber shall be OM3 compliant and should support transmission of 10G signal up to a distance of 300 m for 850 nm fiber and 550 m for 1300 nm fiber.

Comply

- e. Shall contain a Rip Cord applied longitudinally under the cable

jacket for easy cable jacket removal.

- f. Shall contain a lightweight Central Strength member located in the middle of the fiber bundles. Comply
- g. Shall contain both color-coded buffered fibers as well as color-coded buffer tubes. Comply
- h. Cables shall have length markings in 2 ft. increments. Comply
- i. Ethernet application support:

Application	Distance (m)
10GBASE-SX (850 nm)	300
10GBASE-LX4 (1300 nm)	300
1000BASE-SX (850 nm)	900
1000BASE-LX (1300 nm)	600
Fibre Channel 266 (1300 nm)	1,500
ATM 622 (1300 nm)	500
ATM 155 (1300 nm)	2,000
ATM 52 (1300 nm)	3,000
FDDI (Original 1300 nm)	2,000
100BASE-FX (1300 nm)	2,000

- j. Shall meet the minimum performance parameters: Comply

Minimum Performance Parameters for Qualified Cables					
Minimum Bandwidth (MHz-km)		Maximum Attenuation (dB/km)		Group Index of Refraction	
850 nm	1300 nm	850 nm	1300 nm	850 nm	1300 nm
Laser - 2000 OFL - 1500	OFL - 500	3.0	1.0	1.483	1.479

- k. Physical Characteristics: Comply

Core Size (Microns)	Cladding Size (Microns)	Coating Size (Microns)	Buffer Size (Microns)	Core Cladding Concentricity (Microns)
50 ± 3	125 ± 2	245 ± 10	900 ± 50	≤3.0

3. Single Mode Fiber Optic Cable

- a. The Cabling system must be designed and installed according to ISO 11801 OS1 compliant and should also meet EN50173 2nd editions and TIA/EIA 568B3. The Cable shall have 6 or 12 or 24 cores, 9.2/125 µm universal distribution cable with improved performance. Comply
- b. Low Smoke Zero Halogen - LSZH Jacket that does not give off

toxic fumes in case of fire and offer flame propagation retardant properties. **Comply**

- c. Shall contain a Rip Cord applied longitudinally under the cable jacket for easy cable jacket removal. **Comply**
- d. Shall contain a lightweight Central Strength member located in the middle of the fibre bundles. **Comply**
- e. Shall contain both color-coded buffered fibers as well as color-coded buffer tubes. **Comply**
- f. Cables shall have length markings in 2 ft. increments.
- g. Fiber will be available in strand counts of 4, 6, 12, 24, 36, 48, 72, 96, 144 and 288.
- h. Shall meet these minimum performance parameters:

Minimum Performance Parameters for Qualified Cables							
Fiber	Cable Type	Maximum Attenuation (dB/km)		Zero dispersion		Index of Refraction	
		1310 nm	1550 nm	Wavelength (nm)	Slope (nm ² -km)	1310 nm	1550 nm
Singlemode	Indoor/Outdoor	0.40	0.30	1302-1322	<0.089	1.468	1.468

- i. Physical Characteristics:

Mode and Fiber Type	Core Size (Microns)	Cladding Size (Microns)	Coating Size (Microns)	Buffer Size (Microns)	Core/Cladding Concentricity (Microns)
Singlemode	8.3 ± 1.0	125 ± 1.0	245 ± 10	900 ± 50	≤ 0.8

- j. Ethernet application support:

Application	Distance (m)
10GBASE-L (1310 nm)	8,000
10GBASE-E (1550 nm)	30,000
10G Fiber Channel (Serial-1310 nm)	10,000
10G Fiber Channel (WDM-1310 nm)	10,000
1000BASE-LX (1300 nm)	5,000
Fiber Channel 266/1062 (1300 nm)	10,000
ATM 52/155/622 (1300 nm)	15,000

- 4. Optical Fiber Patch Panel: **Comply**

- a. Optical fiber Patch Panels shall be mounted in 19" frames of the cabinets. The patch panels shall be equipped with a mechanism that ensures the retention and support of incoming

cables. An Earth Key shall be provided within the patch panel to earth any metallic part of the cable. The patch panel shall be designed with a sliding mechanism enabling front side installation and maintenance work to be carried out without having to remove the entire panel. **Comply**

- b. The patch panel shall provide facilities to recess the front connector plate deeper than the front of the 19" rails of the cabinet. This will provide sufficient bend radius for the patch cords once connected to the panel. This shall also prevent damage to the patch cords when the cabinet doors are closed. **Comply**

- c. Direct Termination of the connectors on to the fibers as well as splicing of pigtails shall be possible. The Patch Panel shall provide management for 1m of fiber per link after breaking the fibers out from the cable. The Patch Panel shall support the connector type specified for this installation. For multimode fiber links, these can be LC, SC and ST. The panel-mounted couplers shall be protected on the front presentation side of the patch Panel for safety purposes. **Comply**

5. Optical Fiber Patch Cords **Comply with note: Fiber patch cords are provided in 1m and 3m lengths**

- a. The Fiber adapters will be connected to the active equipment by means of duplex patch cords 2LC-2LC, 2 SC-2 SC or 2 ST-2 ST as per client site standard. **Comply**
- b. The patch cords consist of 50/125 microns fibers and an LSHF-FR jacket. The SC or ST connectors shall comply with the International standard IEC 74-13. The patch cords should be available in lengths of 2 and 5 meters. **Comply**
- c. To avoid mix of patch cords built around different types of fiber, cords produced with LASER-optimized fiber will be used for both OM1 and OM2 optical fiber cables. When using OM3 optical fiber cables, patch cords produced with the same OM3 fiber have to be installed. **Comply**
- d. Fiber cables shall interface and connect to fiber interface unit at both ends as part of the fiber contractor work. **Comply**

2.5 CABINETS: **Comply with note: Floor Cabinets 1000mm depth is provided instead of 800mm**

- A. The metal cabinets shall have a footprint of 800x800 mm. In the frames 19" components can be mounted by means of the standard cage nuts. The front door consists of a metal framework with hinges and a central security glass. The side panels and the panel in the back have to be equipped with a hinge on the left or the right in order to facilitate access to the equipment. A 42 U cabinet is preferred, providing enough space for active equipment.
- B. For an orderly cord storage and easy to manage installation, the following accessories shall be used:
 - Closed 1 or 2 U patch guides between the patch's panels.

- Lateral cable rings installed at both sides of the frames. The patch rings can be removed very easily by rotation and have to be located on the front rails of the 19" frames in the cabinets.

The cabinets should be supplied with:

- A power bar with 6 * 240VAC electrical sockets and a mains isolator switch **Depends On Devices Quantities And Load**
- Up to 4 fans (240VAC) for expelling air, approximately 220W each.

Comply as noted That 2 FAN Proposed

C. Earthing has to be achieved by means of the 19" frame attachments, thus ensuring continuity. The cabinet and frame assembly when installed will also serve as equi-potential plane so that damaging external EMI currents can be drained off. To this end, the inter-cabinet connections shall also be made off by extending the earth connection from cabinet to cabinet in a suite of cabinets. The suite of cabinets shall be connected to the grounding network of the building. **Noted**

D. The Earth key of the cabinet must be connected to the protective earth. The dimension of the earth conductor is 6mm². If no or only a poor protective earth system is present in the building, a separate earthing to the main earth terminal of the building is required. The dimensions of the conductor should then be 16mm². **Noted**

For Wall-Mounted Racks: 15U is provided instead of 27U as per BOQ

E. Wall-Mounted Racks:

1. Height: as indicated on Drawings.
2. Depth: as indicated on Drawings
3. Number of Rack Units per Rack: 27.
4. Threads: Universal square.
5. Wall Attachment: Four mounting holes.
6. Equipment Access: Integral swing.

F. Cabinet Mounted UPS (Uninterruptible Power Supply): To enable Power Over Ethernet (POE) a smart UPS shall be supplied and installed as a part of the cabinet. The UPS shall have the following minimum features:

1. Output **Comply**
 - a. Output Power Capacity: 2250 Watts/3000 VA
 - b. Max. Configurable Power: 2250 Watt/3000 VA
 - c. Nominal Output Voltage: 230 V or as indicated on power drawings.
 - d. Typical Back Time: At full load 5.7 minutes (2250 Watts)
2. Communications & Management **Comply**
 - a. Interface Port(s): DB-9 RS-232, Smart Slot
 - b. Control Panel: LED status display with load and battery bar-